

Where deep learning losses come from II

MLE, regression and classification: each loss is the negative log-score of a predictive distribution.

THE QUESTION

Why is squared error the loss for regression, and cross-entropy the loss for classification?

THE INTUITION

Assume a distribution for the target and write its negative log-likelihood. Gaussian noise around a line gives squared error; a Bernoulli label gives binary cross-entropy; a categorical label gives softmax cross-entropy.

KEY TAKEAWAY

The distribution you assume decides the loss you get. MSE, BCE and CE are all $-\log p(\text{observed target})$.

distribution assumption $\rightarrow$ $-\log p$ per example $\rightarrow$ drop constants $\rightarrow$ MSE · BCE · CE $\rightarrow$ + prior $\rightarrow$ MAP

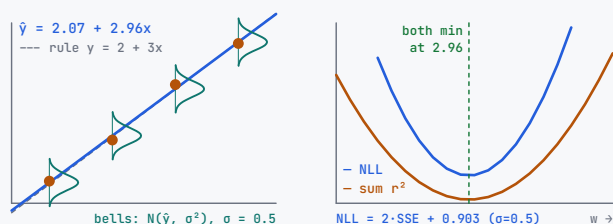
1 The universal derivation

Conditional independence gives $p(\mathcal{D}|\theta) = \prod_i p_\theta(y_i|x_i)$. Take logs and turn maximization into minimization:

$$\hat{\theta}_{\text{MLE}} = \arg \min_{\theta} - \sum_i \log p_\theta(y_i|x_i).$$

Define the one-example loss $\ell_i(\theta) = -\log p_\theta(y_i|x_i)$. The dataset objective is the sum or the mean of these contributions.

2 Gaussian likelihood gives squared error



Assume $Y_i|x_i \sim \mathcal{N}(\mu_i, \sigma^2)$ with $\mu_i = f_\theta(x_i)$. Then

$$-\log p(y_i|x_i) = \frac{1}{2\sigma^2}(y_i - \mu_i)^2 + \log \sigma + C.$$

With σ fixed, $\hat{\theta}_{\text{MLE}} = \arg \min_{\theta} \sum_i (y_i - f_\theta(x_i))^2$. On the four points above, the line (0, 1) scores sum $r^2 = 56.58$, the true rule (2, 3) scores 0.180, and the MLE (2.07, 2.96) scores 0.162.

MSE is Gaussian NLL up to a positive scale and additive constants. Learning σ requires keeping the $\log \sigma$ term.

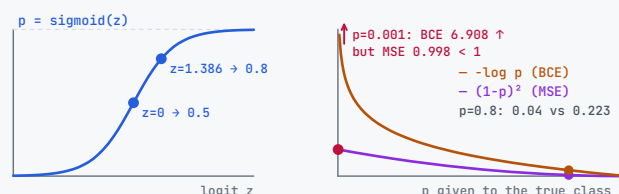
3 The residual model chooses the penalty

With residual $r = y - \hat{y}$:

	noise	NLL in r	effect
Gaussian	r^2		large errors dominate
Laplace	$ r $		constant influence
Student- t	$\log(1 + r^2/(\nu s^2))$		heavy-tail robustness
Uniform	0 in, ∞ out		hard support

Constants and positive scale factors are omitted. Lecture 3 plots these shapes on data with outliers.

4 Bernoulli likelihood gives BCE



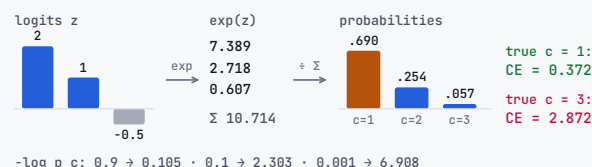
For $y \in \{0, 1\}$ and predicted $p = P(Y = 1|x)$: $p(y|x) = p^y(1-p)^{1-y}$, so

$$\ell_{\text{BCE}} = -[y \log p + (1-y) \log(1-p)].$$

If $y = 1$, $p = 0.8$ gives $\ell \approx 0.22$ while $p = 0.2$ gives $\ell \approx 1.61$. MSE is possible but mismatched: $(1-p)^2$ never exceeds 1, while $-\log p$ is unbounded.

Output a logit $z \in \mathbb{R}$ and use a fused BCE-with-logits operation for numerical stability.

5 Categorical likelihood gives cross-entropy



For mutually exclusive classes, logits z define $p_k = \exp z_k / \sum_j \exp z_j$. If the observed class is c ,

$$\ell_{\text{CE}} = -\log p_c = -z_c + \log \sum_j \exp z_j.$$

CE depends only on the probability given to the true class. Confident mistakes are expensive:

$-\log p_c \rightarrow \infty$ as $p_c \rightarrow 0$.

6 Match observation, output and loss

target	model output	NLL
real	μ ; optionally $\sigma > 0$	Gaussian / robust
binary	one unconstrained logit	Bernoulli BCE
one of K	K unconstrained logits	categorical CE
count	positive rate λ	Poisson NLL

Before optimizing a loss, say which random variable and which distribution make it the right score.

From MLE to MAP and regularization

Likelihood asks for data fit; a prior adds an explicit preference among plausible parameter settings.

7 Bayes combines data and prior

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})}$$

likelihood how the fixed data score each θ
prior pre-data preference over θ
posterior updated distribution after seeing the data
evidence normalizer; independent of θ

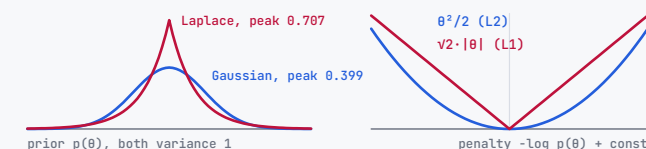
8 MAP is loss plus regularizer

$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta|\mathcal{D})$. Drop the evidence and take negative logs:

$$\hat{\theta}_{\text{MAP}} = \arg \min_{\theta} \left[\underbrace{-\log p(\mathcal{D}|\theta)}_{\text{NLL}} + \underbrace{-\log p(\theta)}_{\text{prior penalty}} \right].$$

MLE uses only the likelihood. **MAP** returns one parameter vector, at the posterior mode.

9 Gaussian prior gives L_2



If $\theta \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$, then $-\log p(\theta) = \frac{1}{2\tau^2} \|\theta\|_2^2 + C$, so

$$\mathcal{L}_{\text{MAP}} = \text{NLL} + \lambda \|\theta\|_2^2, \quad \lambda = \frac{1}{2\tau^2}.$$

Smaller τ : a narrower prior, larger λ , stronger shrinkage toward zero.

10 Laplace prior gives L_1

For independent zero-centred Laplace parameters with scale a , $-\log p(\theta) = \frac{1}{a} \|\theta\|_1 + C$. The corner at zero can set coefficients **exactly** to zero: L_2 shrinks smoothly, L_1 can create sparsity.

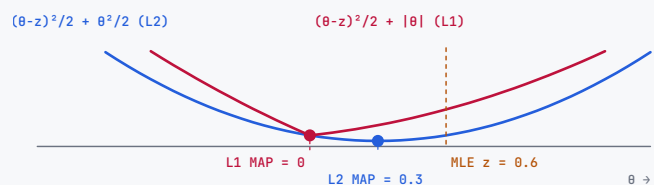
11 Scaling: keep the statistical meaning

The exact MAP objective uses summed NLL plus the full prior penalty: $\sum_i \ell_i(\theta) + \lambda_{\text{sum}} R(\theta)$. If code uses the mean NLL, divide the whole objective by n :

$$\frac{1}{n} \sum_i \ell_i(\theta) + \lambda_{\text{mean}} R(\theta), \quad \lambda_{\text{mean}} = \frac{\lambda_{\text{sum}}}{n}.$$

Changing the batch size should not silently change the intended regularization strength. Document whether a library returns a sum or a mean.

12 One-dimensional MAP shrinkage

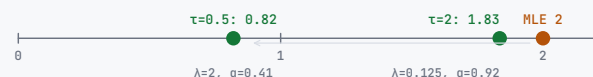


Approximate the likelihood by $p(\mathcal{D}|\theta) \propto \exp(-(\theta - z)^2/(2s^2))$ with $z = \hat{\theta}_{\text{MLE}}$, and take $\theta \sim \mathcal{N}(0, \tau^2)$. Minimizing the two quadratics gives

$$\hat{\theta}_{\text{MAP}} = \underbrace{\frac{\tau^2}{\tau^2 + s^2}}_{\alpha \in (0,1)} z.$$

If $z = 0.6$ and $s = \tau$, then $\hat{\theta}_{\text{MAP}} = 0.3$. With the L_1 penalty $|\theta|$ instead, the minimizer is exactly 0, because $|z| < 1$.

13 What stronger priors do



More data likelihood sharpens; MAP often approaches MLE
Narrower prior posterior mode moves farther toward preferred values (above: MLE 2, $s = 0.6$)
Correlated prior preferred directions become elliptical, not axis-aligned
Prior centre m shrinkage is toward m , not automatically toward zero

14 Final audit

- ✓ Each loss is tied to an explicit likelihood.
- ✓ Constants are dropped only when they do not affect the optimized parameters.
- ✓ Logits and constrained distribution parameters are not confused.
- ✓ Prior family and scale are stated; sum-versus-mean scaling is explicit.

15 In code: weight decay is a Gaussian prior

```
nll = F.mse_loss(pred, y)      # mean NLL
pen = sum((w**2).sum()
           for w in model.parameters())
loss = nll + lam * pen          # MAP (lambda_mean)
# same gradient: SGD(..., weight_decay=2*lam)
```

Plain SGD's `weight_decay = 2λ` adds the gradient of $\lambda \|\theta\|_2^2$. AdamW's decoupled decay is not exactly this penalty.

Likelihood gives the data loss · the prior gives the regularizer · their posterior mode gives MAP

Page 1 numbers from the Lecture 2 storyboard data (four points from $y = 2 + 3x$; logits (2, 1, -0.5)). Page 2 figures use the Lecture 3 storyboard's prior examples. Displayed values rounded. Slides, notebooks and summary videos: nipunbatra.github.io/dl-2026.